SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course Code:	CSA1715	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Application Based Questions, In-class
Weightage:	_____ 40% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
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Section / Batch:	B-tech AI/DS	**Date:**	20-10-2026

Code of Conduct

I, Vaibhava Lakshmi K- (192511310), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:VAIBHAVA LAKSHMI K-

Instructions

- Read each question carefully before answering (in-Class Assessment)
- Each question is a real-world application scenario. Identify the AI technique that fits, apply it step by step, and justify every decision.
- Questions are drawn from Informed Search (Greedy, A*, heuristics), Local Search and Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Show intermediate tables (open lists, domains, α/β values) and state the final answer clearly.
- Marks will be awarded using the Application Based Questions rubric: understanding of problem context, application of concepts, problem-solving approach, accuracy of solution and clarity.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions (admissibility, consistency)	**Q1**
2. Local Search, Online Search & Constraint Satisfaction	Local search and optimization problems, online search agent, CSP (MRV, degree heuristic, forward checking)	**Q2, Q3**
3. Game Playing: Adversarial Search	Games, minimax algorithm, alpha-beta pruning, evaluation functions	**Q4**

Rubrics for Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Problem Context	**20%**	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	**30%**	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	**20%**	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	**20%**	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	**10%**	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (2)	Accuracy of Solution (2)	Clarity (1)	Sub. Total (10)	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A – Application Based Questions

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

Topic: Informed search strategies, heuristic functions | 15 minutes

A campus delivery robot must carry a parcel from the Store (S) to the Hostel (G). The one-way road network and the robot's heuristic estimate h(n) of the remaining travel cost to G (in minutes) are given below.

From	To	Cost
S	A	1
S	B	4
A	C	2
A	D	5
B	G	5
C	D	1
D	G	2

Node	S	A	B	C	D	G
h(n)	6	5	3	3	2	0

Assume ties are broken alphabetically and that a node is re-opened if a cheaper path to it is found.

(a) Apply Greedy Best-First Search from S to G. Write the order in which nodes are expanded, the path returned and its total cost.

(b) Apply A* search. Show the open list with g(n), h(n) and f(n) after each expansion, the order of expansion, and the final path and cost.

(c) Check whether h(n) is admissible and consistent by verifying the required condition for every node and every road.

(d) Compare the outputs of (a) and (b). Explain why Greedy Best-First Search returned a different result and which algorithm you would deploy for the robot, with justification.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

Topic: Local search algorithms and optimization problems, online search agent | 15 minutes

A technician wants to place a Wi-Fi access point at the best position along a corridor with 10 candidate positions x = 0 to 9. The measured signal quality f(x) (higher is better) is:

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

A move shifts the access point to an adjacent position ($x - 1$ or $x + 1$). Steepest-ascent hill climbing moves to the better neighbour and stops when no neighbour is strictly better.

(a) Run steepest-ascent hill climbing from the start positions x = 0, x = 4 and x = 9. Show each step and the final position and signal quality.

(b) Identify all local maxima and the global maximum. If the start position is chosen uniformly at random from the 10 positions, find the probability that one run reaches the global maximum, the probability that at least one of 3 random-restart runs does, and the expected number of runs needed.

(c) A simulated annealing search is at x = 2 and proposes the move to x = 3. Compute the acceptance probability $e^{(\Delta/T)}$ at T = 2 and at T = 1, and explain what this does to the chance of escaping the local maximum as T falls.

(d) The same search is later done by a robot that cannot see the whole corridor, can only measure f(x) where it stands, and cannot undo exploration for free. Explain why this is an online search problem, and state what the agent must remember to avoid cycling.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

Topic: Constraint satisfaction problem (MRV, degree heuristic, forward checking) | 15 minutes

Five exams A, B, C, D and E must each be given one slot from {1, 2, 3}. Two exams that share students must not be in the same slot. The pairs sharing students are: A–B, A–C, A–D, B–C, B–D, C–E, D–E. Because of invigilator availability, exam A can only be in slots 1 or 2, and exam E only in slots 1 or 2.

Exam	Initial domain
A	{1, 2}
B	{1, 2, 3}
C	{1, 2, 3}
D	{1, 2, 3}
E	{1, 2}

Rules for the search: choose the variable using Minimum Remaining Values (MRV), break ties with the degree heuristic, then alphabetically; try values in ascending order; use forward checking after every assignment.

(a) Draw the constraint graph (list its edges) and compute the degree of every exam.

(b) Select the first exam to assign using MRV and the degree heuristic, justify the choice, assign its first value, and show the domains of all exams after forward checking.

(c) Continue the search with the same rules, showing the chosen exam, its value and the updated domains at each step, until a complete assignment is found. Verify it against every constraint.

(d) Interpret: explain how MRV, the degree heuristic and forward checking reduced the search effort here, and what forward checking would signal if a domain became empty.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

Topic: Game playing, adversarial search, evaluation functions | 15 minutes

A game-playing agent (MAX) searches two plies ahead: MAX chooses one of three moves (B, C, D), then MIN replies with one of three moves. Positions at depth 2 are scored by the evaluation function

$$\mathbf{Eval = 2 \times M + 1 \times Mo}$$

where M is the material advantage of MAX and Mo is the mobility advantage of MAX. The leaf features (M, Mo) are given for the leaves in left-to-right order:

MIN node	Leaf 1 (M, Mo)	Leaf 2 (M, Mo)	Leaf 3 (M, Mo)
B	(1, 1)	(5, 2)	(3, 2)
C	(1, 0)	(1, 2)	(2, 2)
D	(6, 2)	(2, 1)	(0, 2)

(a) Compute the Eval value of all nine leaves.

(b) Run plain minimax. Give the backed-up value of B, C and D, the value at the root and the move MAX should play.

(c) Run alpha-beta pruning on the same tree (left to right). Show α and β at each node, mark every pruned branch, and state the root value and the number of leaves evaluated.

(d) If the leaves under C were ordered (6, 4, 2) instead, how many leaves would alpha-beta evaluate? Explain the effect of move ordering and why pruning never changes the minimax value.

ANSWERS – STEP-BY-STEP SOLUTIONS

CSA17 – Artificial Intelligence | CO2 Assessment Tool 3

Question 1: Campus Delivery Robot – Greedy Best-First Search vs A*

(a) Greedy Best-First Search

Greedy Best-First Search selects the frontier node with the smallest heuristic h(n). It ignores the cost already travelled.

Step	Expanded node	Frontier after expansion (ordered by h)
1	S	B (h=3), A (h=5)
2	B	G (h=0), A (h=5)
3	G	Goal reached

Expansion order: S → B → G. Final path: S → B → G. Total cost = cost(S,B) + cost(B,G) = 4 + 5 = 9 minutes.

(b) A* Search

A* evaluates f(n) = g(n) + h(n), where g(n) is the path cost so far. Select the node with the lowest f; break ties alphabetically.

Step	Expanded node	OPEN list after expansion: node (g,h,f)
1	S	A (1,5,6), B (4,3,7)
2	A	C (3,3,6), B (4,3,7), D (6,2,8)
3	C	D (4,2,6), B (4,3,7)
4	D	G (6,0,6), B (4,3,7)
5	G	Goal reached

When C is expanded, the route to D through C costs 3 + 1 = 4, cheaper than the previous cost 6, so D is updated/reopened. Expansion order: S → A → C → D → G. Final path: S → A → C → D → G. Total cost = 1 + 2 + 1 + 2 = 6 minutes.

(c) Admissibility and consistency

Admissibility requires h(n) not to exceed the true shortest cost to G.

Node	h(n)	Actual shortest cost to G	Admissible?
S	6	6	Yes
A	5	5	Yes
B	3	5	Yes
C	3	3	Yes
D	2	2	Yes
G	0	0	Yes

All heuristic values are less than or equal to the actual shortest costs; therefore h is admissible.

Consistency requires $h(u) \le c(u,v) + h(v)$ for every directed road.

Road	Verification	Result
S→A	$6 \le 1+5 = 6$	Yes
S→B	$6 \le 4+3 = 7$	Yes
A→C	$5 \le 2+3 = 5$	Yes
A→D	$5 \le 5+2 = 7$	Yes

B→G	3 ≤ 5+0 = 5	Yes
C→D	3 ≤ 1+2 = 3	Yes
D→G	2 ≤ 2+0 = 2	Yes

Every road satisfies the condition, so the heuristic is consistent.

(d) Comparison and deployment

Feature	Greedy Best-First	A*
Evaluation	h(n)	g(n)+h(n)
Path	S→B→G	S→A→C→D→G
Cost	9 minutes	6 minutes

A* is preferred because it considers both the cost already travelled and the estimated remaining cost, and it finds the lower-cost route in this problem. Greedy search follows the smallest h value and returns a route costing 9 minutes.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

(a) Steepest-ascent hill climbing

Hill climbing moves to a strictly better adjacent position and stops when neither neighbour has a higher value.

Start	Steps (x : f(x))	Final position
x=0	0:2 → 1:5 → 2:9	x=2, f(x)=9
x=4	4:4 → 5:7 → 6:12	x=6, f(x)=12
x=9	9:3 → 8:8 → 7:10 → 6:12	x=6, f(x)=12

(b) Maxima and probabilities

Local maxima: x=2 with signal 9, and x=6 with signal 12. The global maximum is x=6, f(x)=12.

Starting positions that reach the global maximum are 4, 5, 6, 7, 8 and 9: 6 of 10 positions. Thus p = 6/10 = 0.6.

Probability that at least one of three independent random-restart runs succeeds = $1 - (1-p)^3 = 1 - (0.4)^3 = 0.936 = 93.6\%$.

Expected number of runs until success = 1/p = 1/0.6 ≈ 1.67 runs.

(c) Simulated annealing

The move is x=2 to x=3. Δ = f(3)−f(2) = 6−9 = −3. For a worse move, P = e^(Δ/T).

At T=2: P=e^(−3/2)=e^(−1.5)≈0.2231=22.31%. At T=1: P=e^(−3)≈0.0498=4.98%.

As temperature decreases, acceptance of worse moves becomes less likely. Higher temperature allows more exploration and can help escape a local maximum; lower temperature makes the search more selective.

(d) Online search

It is online search because the robot cannot observe the entire corridor in advance and can measure signal quality only at its current position. To avoid cycling, it should remember its current position, previously visited positions, measured signal values, and actions or paths already attempted.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

(a) Constraint graph and degrees

Variables are exams A, B, C, D and E; domains are available time slots. Exams sharing students must have different slots. Edges: A–B, A–C, A–D, B–C, B–D, C–E, D–E.

Exam	Connected exams	Degree
A	B, C, D	3

B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

(b) First variable using MRV and degree heuristic

MRV chooses the variable with the fewest remaining values. A and E each have two values. Their degrees are 3 and 2 respectively, so the degree heuristic selects A. Try the lowest value: A=1.

Exam	Domain after assigning A=1
A	{1}
B	{2,3}
C	{2,3}
D	{2,3}
E	{1,2}

Forward checking removes slot 1 from B, C and D. No domain becomes empty.

(c) Continue the search

Step 1: A=1. Domains: B={2,3}, C={2,3}, D={2,3}, E={1,2}.

Step 2: B, C and D tie on domain size and degree; choose B alphabetically. Assign B=2. Forward checking removes 2 from C and D, leaving C={3}, D={3}, E={1,2}.

Step 3: C and D have one value each and tie; choose C alphabetically. Assign C=3. E remains {1,2}.

Step 4: Assign D=3. E remains {1,2}, since E is not connected to D's slot 3? The constraint D–E means E must differ from D; slots 1 or 2 are both valid.

Step 5: Assign E=1, the lowest remaining value.

Exam	Assigned slot
A	1
B	2
C	3
D	3
E	1

Constraint edge	Slots	Valid?
A–B	1 and 2	Yes
A–C	1 and 3	Yes
A–D	1 and 3	Yes
B–C	2 and 3	Yes
B–D	2 and 3	Yes
C–E	3 and 1	Yes
D–E	3 and 1	Yes

(d) How CSP techniques reduce search

MRV assigns the most constrained exam first, revealing conflicts early. The degree heuristic prioritizes the exam affecting the most unassigned exams. Forward checking removes incompatible values from neighbours after each assignment. If a domain becomes empty, the current assignment cannot lead to a solution and backtracking is required.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation

(a) Evaluation values of all nine leaves

Evaluation function: Eval = 2 × M + Mo.

MIN node	Leaf	M	Mo	Calculation	Eval
B	B1	1	1	2×1+1	3
B	B2	5	2	2×5+2	12
B	B3	3	2	2×3+2	8
C	C1	1	0	2×1+0	2
C	C2	1	2	2×1+2	4
C	C3	2	2	2×2+2	6
D	D1	6	2	2×6+2	14
D	D2	2	1	2×2+1	5
D	D3	0	2	2×0+2	2

(b) Plain minimax

MIN selects the lowest child value at each node: B=min(3,12,8)=3; C=min(2,4,6)=2; D=min(14,5,2)=2. MAX selects the largest of these: max(3,2,2)=3. Root value=3; MAX should choose B.

(c) Alpha-beta pruning, left to right

Initially $\alpha=-\infty$ and $\beta=+\infty$. Evaluate B leaves 3, 12, 8; MIN at B returns 3, so root α becomes 3. At C, first leaf is 2, so $\beta=2$. Since $\beta \le \alpha$ ($2 \le 3$), prune C2=4 and C3=6. C returns 2. At D, evaluate 14, then 5, then 2; its value is 2 and the cutoff occurs only after the last leaf, so no leaves under D remain to prune.

Node	Leaves evaluated	Result
B	3, 12, 8	B=3
C	2; prune 4 and 6	C=2
D	14, 5, 2	D=2

Root value=3; best move=B. Leaves evaluated=3+1+3=7. Leaves pruned=2.

(d) Reorder C leaves as (6, 4, 2)

After B, $\alpha=3$. At C, evaluate 6 ($\beta=6>\alpha$), then 4 ($\beta=4>\alpha$), then 2 ($\beta=2 \le \alpha$). The cutoff occurs on the last leaf, so no remaining C leaves can be pruned. All leaves are evaluated: 3 under B + 3 under C + 3 under D = 9. Move ordering affects how early pruning occurs, but alpha-beta pruning does not change the minimax value because it only skips branches that cannot affect the final decision.

Final Revision Summary

Question	Technique	Final result
Q1	Greedy Best-First and A*	Greedy cost=9; A* cost=6
Q2	Hill climbing and simulated annealing	Global maximum x=6; signal=12
Q3	CSP, MRV and forward checking	A=1, B=2, C=3, D=3, E=1
Q4	Minimax and alpha-beta pruning	Root=3; choose B; 7 leaves evaluated